



E.4 Fission

Nuclear and quantum physics · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Nuclear fission is the splitting of a heavy, unstable nucleus (such as ^{235}U) into two smaller daughter nuclei, typically induced by absorption of a slow (thermal) neutron, releasing further neutrons, gamma radiation, and a large amount of energy. This energy release occurs because the daughter nuclei have a higher average binding energy per nucleon than the original heavy nucleus, so the products are more tightly bound and mass is converted to energy via $E = \Delta mc^2$, where Δm is the mass defect between reactants and products. In a nuclear reactor, a self-sustaining chain reaction is maintained at a controlled, steady rate (criticality) using a moderator to slow neutrons (increasing capture probability by fissile nuclei) and control rods (typically boron or cadmium) to absorb excess neutrons and regulate the reaction rate.

Worked example: follow the reasoning

A fission reaction releases a mass defect of $3.20 \times 10^{-28} \text{ kg}$ per nucleus. Find the energy released per fission event. ($c = 3.00 \times 10^8 \text{ m}\cdot\text{s}^{-1}$)

$$E = \Delta mc^2 = (3.20 \times 10^{-28})(3.00 \times 10^8)^2 = (3.20 \times 10^{-28})(9.00 \times 10^{16}) = 2.88 \times 10^{-11} \text{ J}.$$

Answer: $E \approx 2.88 \times 10^{-11} \text{ J}$ per fission ($\approx 180 \text{ MeV}$).

Quick check · multiple choice

In a nuclear reactor, what is the main purpose of the moderator?

- A To absorb excess neutrons and stop the chain reaction
- B To slow down fast neutrons so they are more likely to cause further fission
- C To speed up neutrons for higher energy release
- D To shield workers from gamma radiation

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A single fission event of ^{235}U releases approximately $3.2 \times 10^{-11} \text{ J}$ of energy. [7]
- (a) Calculate the energy released, in joules, by the fission of 1.0 mol of ^{235}U nuclei (Avogadro's number $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$).
 - (b) A reactor using this fuel needs to generate a continuous power output of 800 MW . Calculate the number of fission events required per second.
 - (c) Explain, in terms of control rods, how the reactor is kept at a stable, controlled power output (criticality) rather than an increasing chain reaction.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $E_{\text{total}} = (3.2 \times 10^{-11})(6.02 \times 10^{23}) = 1.926 \times 10^{13} \text{ J}$.

(b) Number per second = $P/E_{\text{per fission}} = (800 \times 10^6)/(3.2 \times 10^{-11}) = 2.5 \times 10^{19}$ fissions per second.

(c) Control rods made of neutron-absorbing material (e.g. boron or cadmium) are inserted to varying depths into the reactor core; by absorbing a proportion of the free neutrons, they ensure on average exactly one neutron from each fission goes on to cause another fission (criticality, $k=1$), keeping the reaction rate and power output steady rather than growing exponentially.

Quick check answer

B. The moderator (e.g. water or graphite) slows fast neutrons released in fission to thermal speeds, since slow neutrons are far more likely to be captured by fissile nuclei (like ^{235}U) and induce further fission, sustaining the chain reaction.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.